

MARKYTICS CONSULTING PVT LTD

# Warehouse & Logistics Management System

Intern Project Task Sheet

|                                |                               |                               |
|--------------------------------|-------------------------------|-------------------------------|
| <b>DURATION</b><br>1 – 2 Weeks | <b>DIFFICULTY</b><br>Advanced | <b>DOMAIN</b><br>Supply Chain |
|--------------------------------|-------------------------------|-------------------------------|

## Objective

Develop a Warehouse and Logistics Management System that helps businesses manage stock storage, inbound/outbound shipments, and delivery tracking. The platform should streamline warehouse operations by automating inventory updates, shipment scheduling, and generating logistics reports.

## Core Features

|          |                                                                                                                                                                                                                                                                         |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b> | <b>Authentication &amp; Roles</b> <ul style="list-style-type: none"><li>– Role-based access for Admin, Warehouse Manager, and Delivery Agent.</li><li>– Secure login and profile management.</li></ul>                                                                  |
| <b>2</b> | <b>Stock &amp; Storage Management</b> <ul style="list-style-type: none"><li>– Add and manage stock items with: SKU, Name, Category, Quantity, Weight, and Warehouse Zone.</li><li>– Real-time stock level tracking with low-stock alerts.</li></ul>                     |
| <b>3</b> | <b>Inbound &amp; Outbound Shipments</b> <ul style="list-style-type: none"><li>– Log inbound shipments (GRN – Goods Receipt Notes) against purchase orders.</li><li>– Create outbound dispatch orders with item list, destination, and expected delivery date.</li></ul> |
| <b>4</b> | <b>Delivery Tracking</b> <ul style="list-style-type: none"><li>– Assign delivery agents to dispatch orders.</li><li>– Real-time status updates: Dispatched → In Transit → Delivered / Failed.</li></ul>                                                                 |
| <b>5</b> | <b>Reports &amp; Analytics</b> <ul style="list-style-type: none"><li>– Warehouse occupancy report with zone-wise stock breakdown.</li><li>– Shipment reports: on-time delivery rate, total dispatches, and pending deliveries.</li></ul>                                |

## Tech Stack

|                          |                             |                               |                             |
|--------------------------|-----------------------------|-------------------------------|-----------------------------|
| <b>BACKEND</b><br>Django | <b>FRONTEND</b><br>React.js | <b>DATABASE</b><br>PostgreSQL | <b>DEPLOYMENT</b><br>Docker |
|--------------------------|-----------------------------|-------------------------------|-----------------------------|

|                             |                       |  |                |
|-----------------------------|-----------------------|--|----------------|
| Django Rest Framework (DRF) | Axios<br>React Router |  | docker-compose |
|-----------------------------|-----------------------|--|----------------|

### Deliverables

- ✓ Backend APIs for stock management, shipment workflows, and delivery status tracking.
- ✓ Frontend with warehouse dashboard, shipment forms, and delivery tracking views.
- ✓ Low-stock alert notifications for warehouse managers.
- ✓ PostgreSQL schema for stock, shipments, zones, and delivery agents.
- ✓ Deployment setup using Docker and docker-compose.

### Evaluation Criteria

- Correctness of stock management and shipment workflow logic.
- Real-time tracking implementation and alert mechanisms.
- Dashboard clarity and data accuracy.
- Code quality, modularity, and use of best practices.
- On-time delivery as per the assigned timeline.